# CO1 – Security Features in Real-World Applications

## 1. Introduction

Cryptography plays an important role in keeping information safe when we use applications and online services every day. It helps protect messages, passwords, payment details, and personal data from unauthorized people. Different security features such as confidentiality, integrity, authentication, authorization, and availability use security and cryptographic techniques in different ways. For example, encryption keeps information private, while authentication helps verify that a user is really who they claim to be. Technologies such as TLS, MFA, tokenization, and end-to-end encryption are commonly used in applications like Instagram, Telegram, Google Pay, Netflix, and YouTube. Thus, cryptography and cybersecurity help make online communication and transactions more secure, private, and trustworthy.

## 2. Security Features in Real-World Applications

| Application | Security Feature | Security Mechanism | Protocol / Technology | How it provides security |
|---|---|---|---|---|
| Instagram | Authentication | Multi-Factor Authentication (MFA) | TOTP / Authentication App | A user enters a password and a one-time code. This makes it harder for someone else to access the account, even if the password is known. |
| Telegram | Confidentiality | End-to-End Encryption | MTProto / Secret Chats | Messages are encrypted so that only the intended users can read them. This keeps private conversations confidential. |
| Google Pay | Confidentiality & Integrity | Encryption and Tokenization | TLS / Payment Tokenization | Sensitive payment information is protected using encryption, while tokenization replaces actual card details with a secure token. This helps protect payment data during transactions. |
| Netflix | Authorization | Access Control | OAuth 2.0 / Access Tokens | After logging in, access controls and tokens help determine what resources the user is allowed to access. This prevents unauthorized access to protected accounts and services. |
| YouTube | Availability | Distributed Servers and Load Balancing | CDN / HTTPS | Videos are delivered through multiple servers, helping the service remain available when many users access it at the same time. HTTPS also protects communication between the user and the service. |

## 3. Related Explanations and Definitions

- **Cryptography:** The technique of protecting information by changing it into a form that unauthorized people cannot easily understand.
- **Encryption:** Converts readable data (plaintext) into unreadable data (ciphertext). A key is used to encrypt and decrypt the information.
- **Confidentiality:** Ensures that information can only be viewed by authorized people. Encryption is a common way to provide confidentiality.
- **Integrity:** Ensures that information is not changed or damaged without permission. Cryptographic techniques such as hashes, message authentication codes, and TLS help detect or prevent unwanted changes.
- **Authentication:** Checks whether a user is really who they claim to be. Passwords, MFA, and one-time passwords are common authentication methods.

- **Authorization:** Determines what an authenticated user is allowed to access or do. Access tokens and permissions are commonly used for this purpose.
- **Availability:** Ensures that a service or resource is available when users need it. CDNs, backups, replication, and load balancing help maintain availability.
- **TLS:** Transport Layer Security protects data exchanged between a device and a server by providing encryption and integrity protection.
- **End-to-End Encryption:** Encrypts information so that it can be read only by the intended communicating users or endpoints.
- **Tokenization:** Replaces sensitive information, such as a card number, with a token. The token can be used instead of exposing the original information.
- **MFA:** Multi-Factor Authentication requires two or more types of evidence to verify a user's identity, such as a password and a one-time code.
- **CDN:** A Content Delivery Network uses servers in different locations to deliver content efficiently and help keep online services available.

## 4. How Cryptography Supports Cybersecurity

- It protects sensitive information from unauthorized access.
- It helps secure communication between users and online services.
- It helps protect passwords, payment information, and private messages.
- It supports data integrity by helping detect unauthorized changes to information.
- It works together with authentication and authorization to provide safer access to applications.
- It builds trust in online communication, transactions, and digital services.

## 5. Key Learnings

1. Cryptography is a major part of modern cybersecurity.
2. Different applications use different security mechanisms depending on what they need to protect.
3. Encryption mainly helps provide confidentiality and can also support integrity when used with authentication mechanisms.
4. MFA provides stronger authentication by requiring more than just a password.
5. Tokenization reduces the exposure of sensitive payment information.
6. Authorization controls what an authenticated user is allowed to access.
7. Availability is mainly supported by infrastructure such as CDNs, replication, and load balancing rather than encryption alone.
8. Good cybersecurity usually combines cryptography, authentication, authorization, and reliable infrastructure rather than depending on one technique.

## 6. Conclusion

Cryptography and cybersecurity work together to protect the applications and services we use every day. From logging into Instagram to sending private messages on Telegram and making payments through Google Pay, security mechanisms are used to protect users and their information. Understanding confidentiality, integrity, authentication, authorization, and availability helps us understand how real-world applications stay secure and trustworthy.